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Homework 3

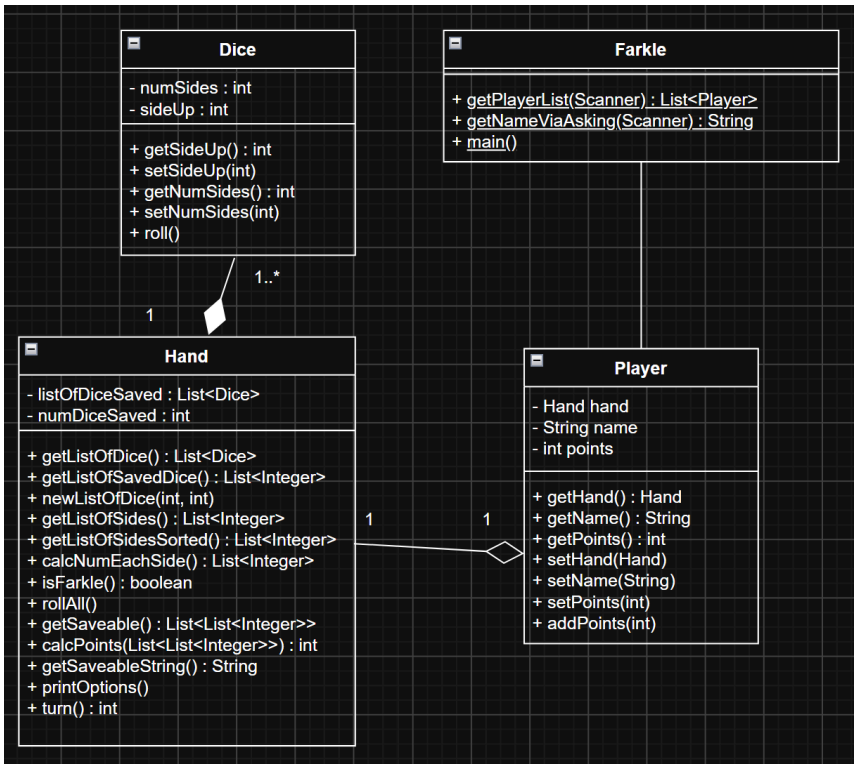
Goal

The goal of this program is to successfully implement the dice game Farkle using a text-based interface. The usual rules of Farkle are applied: a player rolls six dice and each time can choose to set aside certain dice that score. If no dice score, the player scores no points that round and their turn is over. If the player chooses to end before no dice score, they keep the dice they have set aside and add the points to their total points. In this program, one or more players can participate. Each player enters their name, or just hits enter if they want to be assigned a default name.

Overview

My program contains four classes: Dice, Hand, Player, and Farkle. Farkle is where the main work is done, where the user is asked the names and whether the game should be continued. Dice handles, well, the dice and keeps track of which side is currently up on the die. The hand contains a list of saved dice as well as the number of sides the die has. I wanted to make the program as open as possible, so I allowed for a different number of sides than 6 to be allowed. Each Player object has a name, points, and a hand object.

UML Class Diagram



Programming issues

One of the main programming issues I had was implementing the basic rules. Checking for the amount of a certain number of dice was very difficult, and for a long time my program was returning random combinations of numbers. I ended up adding several methods such as calcNumEachSide and getListOfSidesSorted so I could more easily calculate how many dice were on a specific number efficiently. My getSaveable method still ended up being messy, as there are so many checks for how many dice are on a certain side, but it works very well, and I'm ultimately proud of it.

Retrospective

Retroactively, I probably would've created some other class to better organize the players' turns and print all the necessary information for the user.

Appendix

Intro screen:

```
Farkle Rolling and Scoring in Java
**************************************************************************
*                     Zag Farkle by Mac Prouty!                          *
*                         Copyright: 2026                                *
**************************************************************************
```

Full House combo working:

```
Jennifer, it is your turn! Rolling dice...
Here is your roll: [1, 1, 2, 1, 5, 5]
Here are your options for dice to set aside:
A. [1, 1, 1, 5, 5]
```

Single player game:

```
How many players are playing? 1

What is player #1's name? If you would like to be assigned a default name, hit enter
Mac
```

```
Current total points:
Mac 900

Game Over!
Winner: Mac with 900 points!
```

Multiplayer game:

```
How many players are playing? 2

What is player #1's name? If you would like to be assigned a default name, hit enter
Mac
What is player #2's name? If you would like to be assigned a default name, hit enter
Jack

How many points would you like to play in this game? 2000
```

```
Current total points:
Mac 1800
Jack 2200

Game Over!
Winner: Jack with 2200 points!
```